

PAPER 2

1. Introduction

Alzheimer's disease is a progressive neurodegenerative disorder that impairs cognitive functions. Early diagnosis is critical for effective management. In this project, we developed a machine learning model to classify Alzheimer's disease stages using structured patient data.

2. Research Paper Reference

We referred to the research paper "**Machine Learning-Based Prediction of Alzheimer's Disease**", which discusses feature selection, data preprocessing, and model optimization for Alzheimer's classification. The paper influenced our approach in using **Support Vector Machines (SVM)** and **Decision Trees (DT)**.

3. Methodology

3.1 Data Acquisition & Preprocessing

- **Dataset Source:** Kaggle's Alzheimer's Disease Dataset
- **Number of Records:** 2149
- **Number of Features:** 35
- **Preprocessing Steps:**
 - Dropped irrelevant columns (e.g., DoctorInCharge)
 - Handled missing values by replacing them with column means
 - Encoded categorical variables where necessary
 - Standardized numerical features using **StandardScaler**
 - Applied **Principal Component Analysis (PCA)** for dimensionality reduction (10 components retained)

3.2 Model Selection & Training

We implemented the following models:

- **Support Vector Machine (SVM):** Linear kernel, optimized for classification.
- **Decision Tree Classifier:** Used for interpretability and feature importance analysis.

Training Setup:

- **Loss Function:** Cross-Entropy Loss (for multi-class classification)
- **Optimizer:** Default for SVM and Decision Tree
- **Data Split:** 80% Training, 20% Testing

3.3 Evaluation Metrics

To assess model performance, we used:

- **Accuracy Score:** Measures overall correctness.
- **Confusion Matrix:** Visualizes misclassification rates.
- **Precision, Recall, and F1-Score:** Evaluates model effectiveness in detecting Alzheimer's.
- **ROC Curve & AUC Score:** Measures classification confidence.
- **Precision-Recall Curve:** Assesses the trade-off between precision and recall.

4. Results & Analysis

- **SVM Accuracy:** 72.56%
- **Decision Tree Accuracy:** 63.95%
- **SVM Classification Report:**
 - **Precision (Class 0):** 75%
 - **Recall (Class 0):** 86%
 - **F1-Score (Class 0):** 80%
 - **Precision (Class 1):** 66%

- **Recall (Class 1):** 48%
- **F1-Score (Class 1):** 55%
- **Overall Weighted Avg F1-Score:** 71%
- **Decision Tree Classification Report:**
 - **Precision (Class 0):** 73%
 - **Recall (Class 0):** 70%
 - **F1-Score (Class 0):** 72%
 - **Precision (Class 1):** 49%
 - **Recall (Class 1):** 52%
 - **F1-Score (Class 1):** 51%
 - **Overall Weighted Avg F1-Score:** 64%
- **Confusion Matrix Analysis:** Shows lower misclassification for advanced Alzheimer's cases.
- **Precision-Recall Curve:** Indicates strong predictive ability for early-stage detection.

5. Conclusion & Future Work

This project successfully implemented a machine learning model for Alzheimer's disease classification. **Future improvements include:**

- Integrating deep learning approaches like **Neural Networks**.
- Expanding the dataset with MRI-based features.
- Deploying the model using a **web-based interface** for clinical use.

SVM Accuracy: 0.7255813953488373

Decision Tree Accuracy: 0.6395348837209303

SVM Classification Report:

	precision	recall	f1-score	support
0	0.75	0.86	0.80	277
1	0.66	0.48	0.55	153
accuracy			0.73	430
macro avg	0.70	0.67	0.68	430
weighted avg	0.72	0.73	0.71	430

Decision Tree Classification Report:

	precision	recall	f1-score	support
0	0.73	0.70	0.72	277
1	0.49	0.52	0.51	153
accuracy			0.64	430
macro avg	0.61	0.61	0.61	430
weighted avg	0.64	0.64	0.64	430

Models saved successfully!





